

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - IV

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Queue Application:

- a. Write a program to: a. Implement a queue using an array.

Code:

```
import array as arr

que = arr.array('i',[])

si = 6

print(que)

def enqueue(data):
    if(len(que) == si):
        print("Queue Overflow")
    else:
        que.append(data)
        print(que)

def dequeue():
    if(len(que) == 0):
        print("Queue Underflow")
    else:
        que.pop(0)
        print("Data removed: \n",que)

def peek():
    if(len(que) == 0):
        print("Queue is Empty")
    else:
        print("First element is: ", que[0])

def show():
    if(len(que) == 0):
        print("Queue is Empty")
    else:
        print("---- QUEUE ---- \n", que)
```

```
enqueue(1)
```

```
dequeue()
```

```
dequeue()
```

```
enqueue(2)
```

```
enqueue(3)
```

```
enqueue(4)
```

```
enqueue(5)
```

```
enqueue(6)
```

```
enqueue(7)
```

```
enqueue(8)
```

```
dequeue()
```

```
peek()
```

```
dequeue()
```

```
show()
```

```
print("NirajRane_S032")
```

Output:

```
== RESTART: C:/Users/mvlui/OneD  
array('i')  
array('i', [1])  
Data removed:  
array('i')  
Queue Underflow  
array('i', [2])  
array('i', [2, 3])  
array('i', [2, 3, 4])  
array('i', [2, 3, 4, 5])  
array('i', [2, 3, 4, 5, 6])  
array('i', [2, 3, 4, 5, 6, 7])  
Queue Overflow  
Data removed:  
array('i', [3, 4, 5, 6, 7])  
First element is: 3  
Data removed:  
array('i', [4, 5, 6, 7])  
---- QUEUE ----  
array('i', [4, 5, 6, 7])  
NirajRane_So32  
|
```

- b. Simulate a simple queuing system (e.g., customer service queue).

Code:

DataSeturePrac4B.py - C:/Users/mvlui/OneDrive/Documents/SYIT_032/DataStructurePrac4B.py (3.14.6)

File Edit Format Run Options Window Help

```
import time
que = [("C1",0), ("C2",2), ("C3",4), ("C4",6), ("C5",8)]

print(" ----- Customer Order ---|----- ")
print(que)

while que:
    Order, timer = que.pop(0)
    print(f" Serving: {Order} {timer} sec")
    time.sleep(timer)

print(que)

print("NirajRane_So32")
```

Output:

```
===== RESTART: C:/Users/m
----- Customer Order -----
[('C1', 0), ('C2', 2), ('C3', 4), ('C4', 6), ('C5', 8)]
Serving: C1 0 sec
Serving: C2 2 sec
Serving: C3 4 sec
Serving: C4 6 sec
Serving: C5 8 sec
[]
NirajRane_So32
```

Curiosity Questions:

1. Where do you see queues in daily life?
2. Why is **FIFO (First In First Out)** fair in real life? Can you think of situations where it's *not* fair?
3. What will happen if you try to **enqueue more elements than the queue size**?
4. What if you **dequeue from an empty queue**?
5. Imagine you're designing a **hospital emergency system**. Should it use a simple queue or something else? (hint: priority queue).
6. If **5 customers arrive at the same time**, but the service desk can serve only 2 at once, how would you design that?